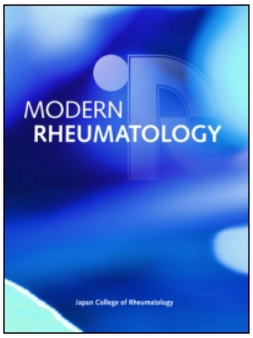

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Modern Rheumatology

ISSN: 1439-7595 (Print) 1439-7609 (Online) Journal homepage: http://www.tandfonline.com/loi/imor20

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To cite this article: Yan-Jie Zhang, Li Zhang, Xiao-Lei Huang, Yu Duan, Li-Juan Yang & Jing Wang (2017): The association between homocysteine and systemic sclerosis: A review of the literature and meta-analysis, Modern Rheumatology, DOI: 10.1080/14397595.2017.1386844

To link to this article: http://dx.doi.org/10.1080/14397595.2017.1386844

Published online: 18 Oct 2017.

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MODERN RHEUMATOLOGY, 2017
https://doi.org/10.1080/14397595.2017.1386844

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ORIGINAL ARTICLE

The association between homocysteine and systemic sclerosis: A review of the literature and meta-analysis

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ABSTRACT

Objectives: The main objective of this study was to summarize the existing evidence and quantitatively evaluate whether serum/plasma levels of homocysteine (Hcy) were associated with sclerosis (SSc) diseases by performing a meta-analysis of previous studies.

Methods: PubMed, Elsevier ScienceDirect and Cochrane Library databases were used to obtain all relative published literatures. Stata version 11.0 (StataCorp, College Station, TX) was used for statistical analysis. The effect size of each study was calculated by the standardized mean difference (SMD) with 95% confidence interval (CI) or quartiles.

Results: A total of eight studies including 475 cases and 265 controls were finally included in this meta-analysis. We found significant between-study heterogeneity and conducted analyses using random-effects models. No significant association was found between the serum levels of Hcy and SSc (pooled SMD = 1.382 μmol/L, 95%CI = −0.442 to 3.206, $p = .137$), but there are two outlier studies that deviate significantly from most other studies, which made it difficult to generalize these results. After excluding these two studies, six studies were included in the meta-analysis. The results showed that the serum levels of Hcy in SSc were significantly higher than that in healthy controls (pooled SMD = 1.182μmol/L, 95%CI = 0.230–2.134, $p = .015$).

Conclusion: Serum/plasma levels of Hcy in SSc diseases were higher than that in healthy controls.

ARTICLE HISTORY
Received 16 May 2017
Accepted 13 September 2017

KEYWORDS
Homocysteine; systemic sclerosis; vascular; Raynaud's phenomenon; fibrosis

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1. Introduction

Sclerosis (SSc) is an autoimmune disorder of the connective tissue, characterized by microvascular damage with associated cutaneous and internal organ fibrosis [1]. Although the clinical expression, severity and progression are rather heterogeneous and it is not clear which of these abnormalities is first expressed, recent studies have known that vascular dysfunction and injury may play a primary role in the pathogenesis of SSc [2]. Moreover, there are several factors that have also been considered to be the possible triggering agents of vascular injury, such as anti-endothelial antibodies, mechanisms of molecular mimicry, microchimerism phenomenon and oxidative stress [3].

Homocysteine (Hcy) is a nonessential sulfur-containing amino acid mainly derived from the diet of methionine, which occurs naturally in the blood plasma. Its chemical properties showed a similarity to cysteine, hence the name homocysteine. In the human body, about 1/2 of Hcy and methyltetrahydrofolate turn into methionine and tetrahydrofolic acid under the action of methylenetetrahydrofolate reductase (MTHFR), and this process needs the help of folic acid and vitamin B12 [4]. The rest of the Hcy turn into cysteine under the catalysis of cystathionine β-synthase (CBS), which require the participation of vitamin B6 (Figure 1). When any factors cause the impair of Hcy metabolism or defects in crucial co-factors that participate in its recycling, it will develop into hyperhomocysteinemia (HHey). There have been a lot of findings shown that HHey was recognized as a modifiable independent risk factor for coronary, cerebral and peripheral vascular diseases [5–8], and experimental studies have also suggested that Hcy induces vascular endothelium injury and leads to vascular damage [3]. These findings suggest that Hcy might be involved in the pathogeneses of both central and peripheral vascular disorders. Recently, many studies have also revealed that multiple rheumatic diseases may be associated with HHey [9–11], and the possible role of HHey in the complex damage mechanisms of systemic rheumatic diseases has also the reason for concern.

In recent years, various similar studies about the association between SSc and Hcy have also been performed (Table 1), but there is controversy between these reports. As shown earlier, several studies have reported that plasma Hcy levels are elevated in SSc patients compared to healthy controls [12–15], but some studies do not support this finding [16–18]. Among them, one study has reported there is no significant difference in Hcy levels between SSc patients and controls, and in another two study performed by Nazarinia

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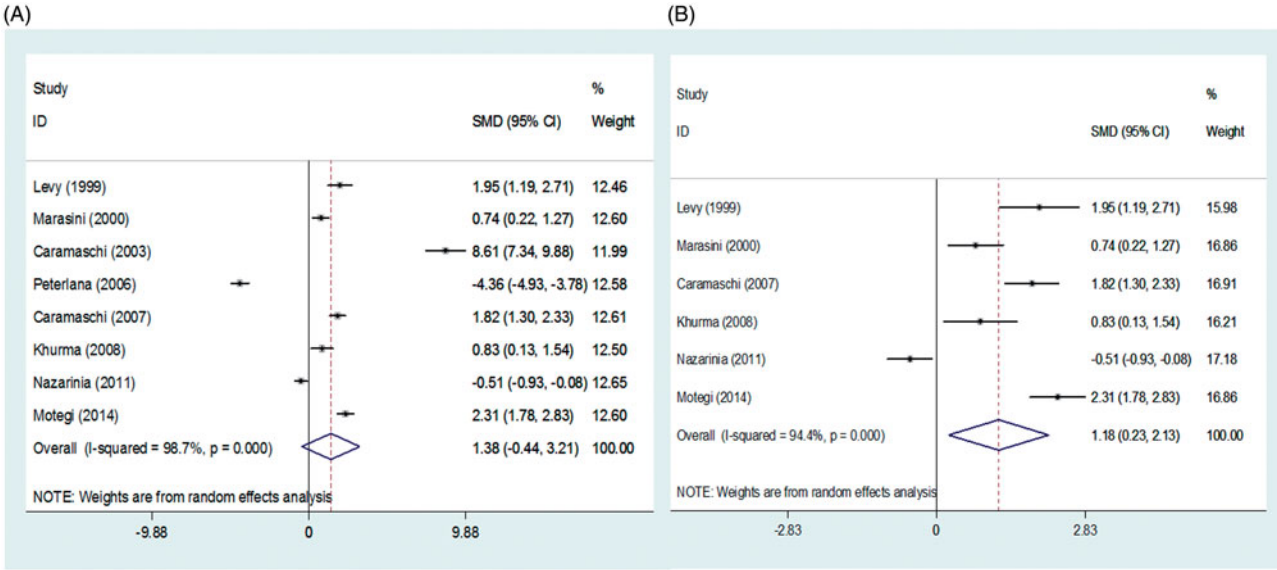

Figure 1. Forest plot of Hcy plasma level and SSc. (A) Meta-analysis of the association between Hcy and SSc; (B) sensitivity analysis (two outlier studies excluded).

Table 1. Association between serum Hcy level and SSc.

First author and year	Samples	Measurement	Serum Hcy levels (μmol/L)	Other findings
Levy 1999 [20]	20/20	HPLC	13.6 ± 5.2 vs. 5.9 ± 2.0	Plasma folate levels lower in patients with primary RP than patients with secondary RP and healthy controls.
Marasini 2000 [21]	30/29	FPIA	13.09 ± 6.53 vs. 9.38 ± 2.53	Patients with SSc had higher Hcy level than patients with primary RP.
Caramaschi 2003 [12]	71/30	HPLC	11.1 ± 0.51 vs. 6.9 ± 0.43	There was a significant association between plasma Hcy concentration and severity of lung involvement.
Peterlana 2006 [18]	82/75	HPLC	11.7 ± 0.58 vs. 14.58 ± 0.74	–
Caramaschi 2007 [13]	60/30	HPLC	11.8 ± 3.1 vs. 6.5 ± 2.5	A significant correlation was found between plasma Hcy concentration and the pattern of NVC with a progressive increase from early to active and above all to late pattern.
Khurma 2008 [14]	17/17	–	10.8 ± 4.4 vs. 8.0 ± 1.8	–
Szamosi 2008 [16]	152/58	HPLC	9.3 vs. 10.1	Higher Hcy concentration was observed in patients with macroangiopathy/thromboembolic events compared to patients without; 45% of patients with HHcy had pulmonary hypertension.
Nazarinia 2011 [17]	44/44	ELISA	19.43 ± 7.20 vs. 22.78 ± 6.01	No statistically significant correlation between serum Hcy level and organ involvement.
Motegi 2014 [15]	151/20	HPLC	9 ± 0.3 vs. 8.2 ± 0.6	Plasma Hcy levels were positively correlated with modified Rodnan total skin score; plasma Hcy levels in SSc with DU, AO and ILD were significantly higher than those in the SSc without.

et al. [17] and Peterlana et al. [18], opposite results have been also observed. These might be explained by small sample sizes or the heterogeneity of the study design, and the methods used to detect plasma Hcy level as well as some inherent biases. Therefore, the aim of the present study was to summarize the available evidence to investigate whether serum/plasma levels of Hcy were associated with SSc using meta-analysis, and summarize recent literature data regarding the reasons for the increased Hcy and the supposed mechanisms involved in SSc.

2. Materials and methods

2.1. Publication search and inclusion criteria

A systematic literature search was performed using the PubMed, Elsevier ScienceDirect and Cochrane Library databases to locate all relevant publications that documented the association of serum levels of Hcy in SSc. Search terms and key words were used as follows: Systemic sclerosis OR scleroderma OR SSc individually with each of the following key descriptors of Hcy: homocysteine, hcy;

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hyperhomocysteinemia, HHcy. Studies meeting the following criteria were considered to be included in this meta-analysis: (a) case–control study, clinical cohort or cross-sectional study design; (b) human subjects with SSc; (c) detailed data about plasma/serum level of Hcy in both patients and healthy controls were accessible. If a study was reported in more than one publication, the latest paper was considered in our meta-analysis.

2.2. Data extraction

All the data were extracted by two independent reviewers according to the inclusion criteria. The extracted information for each eligible papers included the first author's name, publication year, study subjects (cases and controls), assay methods of serum Hcy levels measured, mean Hcy levels (±standard deviation) or median (95%CI/quartiles).

2.3. Statistical analysis

Stata version 11.0 (StataCorp, College Station, TX) was used for statistical analysis. The effect size of each study was calculated by the standardized mean difference (SMD). SMD and 95%CIs were described by a forest plot. In most studies, the mean and deviation were obtained, but in several studies, only the median and 95%CI or quartiles were reported. Therefore, when the original data were median and quartiles, we transformed and calculated the data to gain the appropriate values [19]. I^2 statistics and Q-test were used to assess the degree of heterogeneity between studies. If $p < .05$ for the Q^2 test or $I^2 > 50\%$, a random-effects model was selected to pool the data. Otherwise, a fixed-effect model was used. Begg's test and Egger's test were conducted to evaluate potential publication bias.

3. Results

3.1. Hcy concentration in SSC

We firstly retrieved 31 relevant references from PubMed, Elsevier ScienceDirect and Cochrane Library databases. However, after carefully reading the title and abstract, nine articles [12–18,20,21] with abstracts in total were selected in this systematic review (Table 1). However, in one study [16], only the median was reported. Finally, only eight eligible studies including 475 cases and 265 controls were merged by us. We found significant between-study heterogeneity and conducted analyses using random-effects models. No significant association was found between the serum levels of Hcy and SSc (pooled SMD = 1.382μmol/L, 95%CI = −0.442 to 3.206, p = .137; I^2 = 98.7%, p = .000), but there are two outlier studies [12,18] that deviate significantly from most other studies, which made it difficult to generalize these results. Therefore, we excluded two studies in sensitivity analysis. After excluding these two studies, six studies were included in the meta-analysis. The results showed that the serum levels of Hcy in SSc were higher than that in healthy controls (pooled SMD = 1.182μmol/L, 95%CI = 0.230–2.134, p = .015; I^2 = 94.4%, p = .000] (Figure 1). There was no publication bias performing Egger's regression test (t = 1.20, p = .298) and Begg's test (p = .707) in the six studies (with two outlier studies excluded).

In conclusion, this is the first preliminary meta-analysis to assess the relationship of Hcy serum levels with SSc. However, several limitations should be acknowledged. First, significant heterogeneity was observed in this meta-analysis. The heterogeneity of the population probably originates in the differences in the characteristics of patients, such as age, gender, region, study type and other residual confounding effect. Indeed, the different measurement used to define Hcy plasma level may also play a role in high heterogeneity. Moreover, many factors have an effect on Hcy levels, such as blood folate and vitamin B12 levels and polymorphisms of genes involved in the metabolism of Hcy. Unfortunately, due to the insufficient information of these factors, a comprehensive meta-regression or subgroup analysis was not successfully conducted. Thus, the results of meta-analysis should be interpreted with caution. Therefore, it needs to be demonstrated by well-designed prospective studies whether Hcy was higher in SSc.

3.2. Hcy and clinical features of SSC

As we have shown, many studies have revealed that Hcy plasma level in SSc patients was significantly higher than that in normal controls. And previous studies have already shown that high level of Hcy might be a risk factor for coronary artery disease, stroke and arterial thrombosis in systemic lupus erythematosus (SLE) [22]. Therefore, the correlation between Hcy level and clinical features of SSc has also the reason for concern in recent years (Table 1).

Microvasculopathy is an important pathological changes in SSc. Nailfold videocapillaroscopy (NVC) is a useful tool for the assessment of SSc microvasculopathy. Based on NVC, the microvascular alterations were classified into 'early', 'active' or 'late' pattern [23]. In the study performed by Caramaschi et al., they not only found the plasma Hcy level in SSc was higher than that in controls, but also showed that there was a significant correlation between plasma Hcy concentration and the pattern of NVC with a progressive increase from early to active and above all to late pattern [12]. And in their later study, they further evaluated the association of NVC patterns with both demographic and clinical features in a large group of scleroderma patients, and they still found the Hcy plasma levels were increased across NVC patterns, from early to late pattern in SSc patients [13]. On the other hand, as the patients with SSc typically reveal RP and persistent digital ischemia and occasionally develop digital ulcers (DU), the correlation between Hcy and RP and DU has also already been investigated. For example, both Caramaschi et al. [12] and Marasini et al. [21] have shown a higher Hcy level in SSc patients with secondary RP than healthy controls. In the study performed by Motegi et al., it has found that plasma Hcy levels in the SSc patients with DU were significantly

higher than those in the SSc without DU, despite Hcy level in SSc patients was comparable to that in controls [15]. In conclusion, these above findings suggest that Hcy plasma level may also be related to microvascular involvement and endothelial damage in SSc patients.

In addition to microvascular involvement in SSc, HHey may also play some roles in the pathogenesis of macrovascular damage in SSc. Pulmonary involvement is a common clinical manifestations in SSc, and it mainly includes lung fibrosis and pulmonary hypertension. Szamosi et al. [16] and Caramaschi et al. [12] have investigated the relationship between pulmonary involvement in SSc patients and plasma Hcy levels. A significant association between plasma Hcy concentration and severity of lung involvement has been observed. More interesting, despite there were no significant differences in Hcy levels between SSc patients and healthy controls in Szamosi et al. study [16], higher Hcy concentration was observed in patients with macroangiopathy/thromboembolic events compared to patients without such clinical manifestations and 45% of patients with HHey had pulmonary hypertension. These results suggest that Hcy could be involved in the pathogenetic process of scleroderma pulmonary involvement.

4. Reasons for the increased Hcy in SSc

Hcy's metabolic process is affected by many nutrients (folate and vitamin B12) and various metabolic enzymes (MTHFR and CBS), so HHey is caused either by genetic defects of enzymes involved in Hcy metabolism or by reduced vitamin intake. For example, a previous study suggested that both of serum levels of folate and vitamin B12 were lower in SSc patients than that in controls [24]. This showed that their deficiencies may lead to elevated plasma Hcy level in SSc. However, in addition to dietary vitamin deficiency, genetic defects of enzymes involved in Hcy metabolism may also lead to the rise of plasma Hcy level [16]. Szamosi et al. [16] have found that the frequent mutation (C677T) in the gene encoding for MTHFR may lead to moderate HHey and influence the incidence of macrovascular abnormalities in SSc patients. Beyond that, CBS deficiency may also lead to severe HHey. We all know that homocystinuria is a rare genetic disorder, and it is mainly characterized by HHey that is caused by mutations in CBS, and leads to early-onset atherosclerosis and arterial and venous thromboses [25], and the main treatment strategy to lower cellular and plasma Hcy levels includes vitamin B6 intake, dietary methionine restriction, betaine supplementation, folate and vitamin B12 administration. At the same time, some new strategies to treat homocystinuria have also been made in recent years, and these mainly include functional restoration to mutant CBS, enhanced clearance of Hcy from the body and so on. These suggest that again HHey is mainly provoked by nutritional deficiency and genetic defects of enzymes involved in Hcy metabolism (Figure 2).

However, in addition to the above reasons, other factors may also affect Hcy concentrations, including habitual smoking, coffee intake, some medications, alcohol consumption and physical activity. For instance, a previous study has revealed that the use of diuretics in the SLE group with HHey was significantly higher than that in normohomocysteinemia group (39% vs. 5%; OR = 12.7 [95%CI = 3.3–46.0], $p < .001$) [23]. Moreover, most of the authors have also found that antihypertensive drugs can significantly increase Hcy levels [26,27]. On the other hand, in exploring the relationship between HHcy and lifestyle factors, Chen et al. [28]

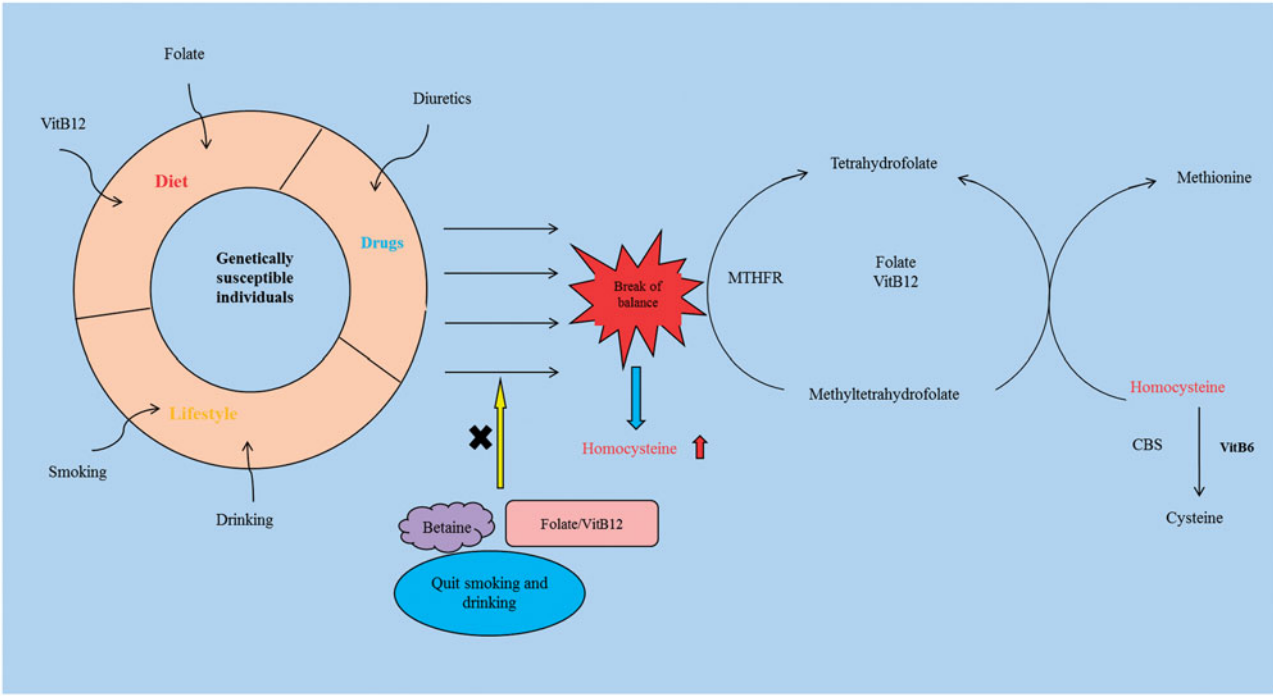

Figure 2. Possible reasons for the increased Hcy in SSc.

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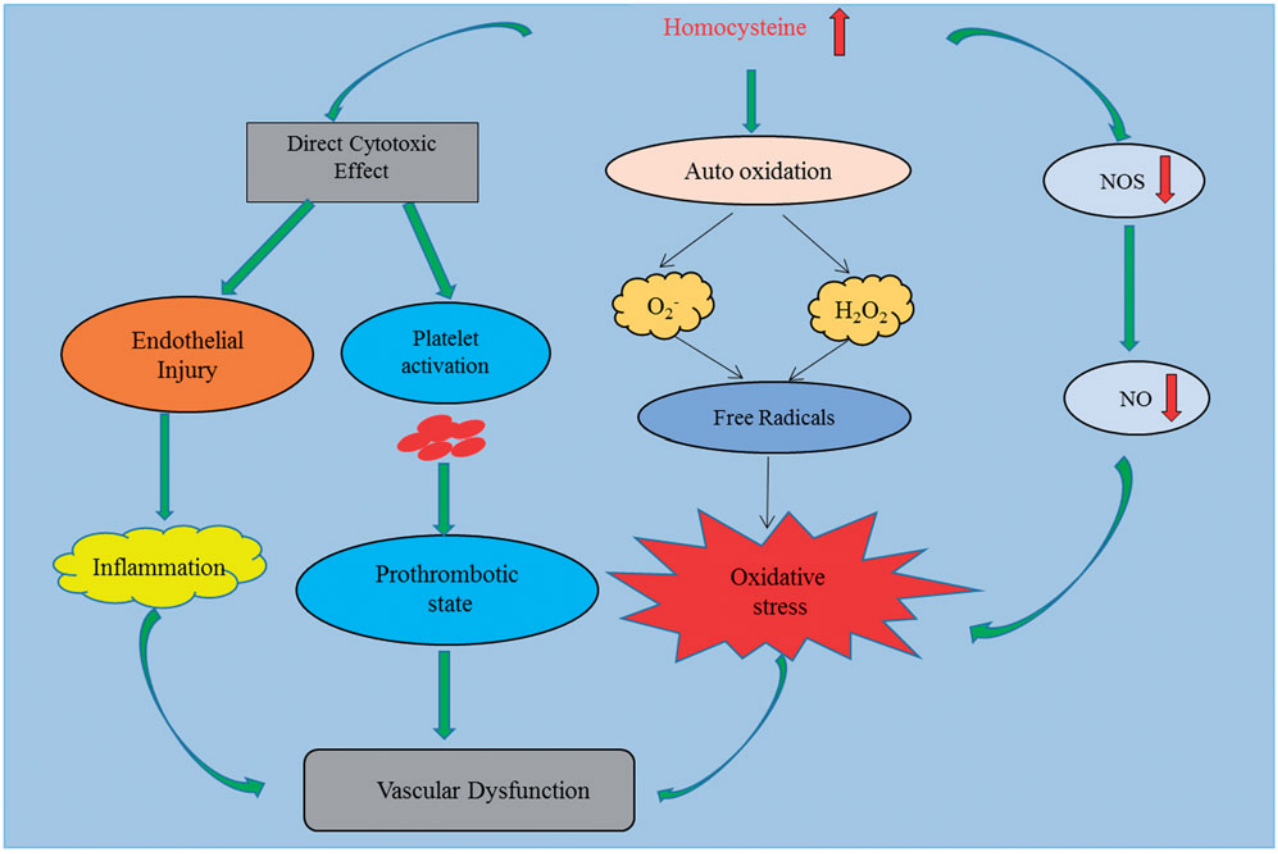

Figure 3. Possible mechanism of Hcy in vascular damage of SSc. Hcy may induce vascular damage through inducing oxidant stress and inflammatory or prothrombotic conditions as well as altering NO release and increasing collagen production. NOS: nitric oxide synthase; NO: nitric oxide.

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have shown that physical activity and current drinking status were significantly associated with HHcy in females, but inversely associated in males. And in both genders, current smoking status and dietary habits were significantly associated with HHey.

In conclusion, a variety of factors have been proved to cause elevated plasma Hcy level. Thus, clinicians should take some corresponding measures to lower patients' plasma Hcy levels, and patients with HHey should also actively adjust their unreasonable diet and lifestyle especially for SSc patients.

5. Possible mechanisms of Hcy in vascular dysfunction of SSc

As shown earlier, high Hcy concentrations have been observed in SSc patients in numerous studies, and high-Hcy levels have also been showed to associate with clinical features of SSc. Currently, many studies have focused on the mechanism of Hcy in vascular dysfunction. As we know, HHey may cause vascular damage by several mechanisms. Firstly, Hcy induces oxidative stress condition. Hcy may favor an oxidant status by generating superoxide radicals. Moreover, Hcy also reported to disrupt uncoupling of nitric oxide (NO) synthase activity, as well as quenching of NO and inhibiting enzyme [3]. Secondly, Hcy induces inflammatory or prothrombotic conditions. Moreover, Hcy can directly induce toxic into endothelium and may influence its physiologic thromboresistance by enhancing platelet activation and enhancing coagulation [29,30]. Finally, it was demonstrated that Hcy alters NO release by cultured endothelial cells and increases collagen production by cultured rabbit smooth muscle cells [31,32] (Figure 3).

Based on the role of Hcy in endothelium dysfunction, the possible Hcy -scleroderma interaction can be explored. Firstly, vascular tone control is abnormal in scleroderma, and vasodilatation and vasoconstriction are imbalance. As we noted above, HHey induces vascular dysfunction through the generation of reactive oxygen species. Moreover, it also leads to the reduction of NO and inactivate NO, which is a powerful regulator of endothelium-dependent vasodilatation [33]. Therefore, it may partially explain the diminished endothelial-dependent vasodilatation in scleroderma by reduction of NO production in endothelial cells. Secondly, many studies have shown that the oxidative stress might play a role in inflammation and fibrosis in heart or liver by using animal models of homocysteinemia [34,35]. At the same time, Raaf et al. [36] have shown that TGF-β levels were elevated in hyperhomocysteinemic rats. These results suggested higher Hcy level is possibly associated with skin sclerosis in patients with SSc. Thirdly, oxygen-free radicals may generate tissue ischemia and subsequently mediate the endothelial injury. Previous studies have suggested that reactive oxygen species may fragment autoantigens, supporting the generation of the typical autoantibodies found in patients with SSc [37]. The ability of Hcy to favor an

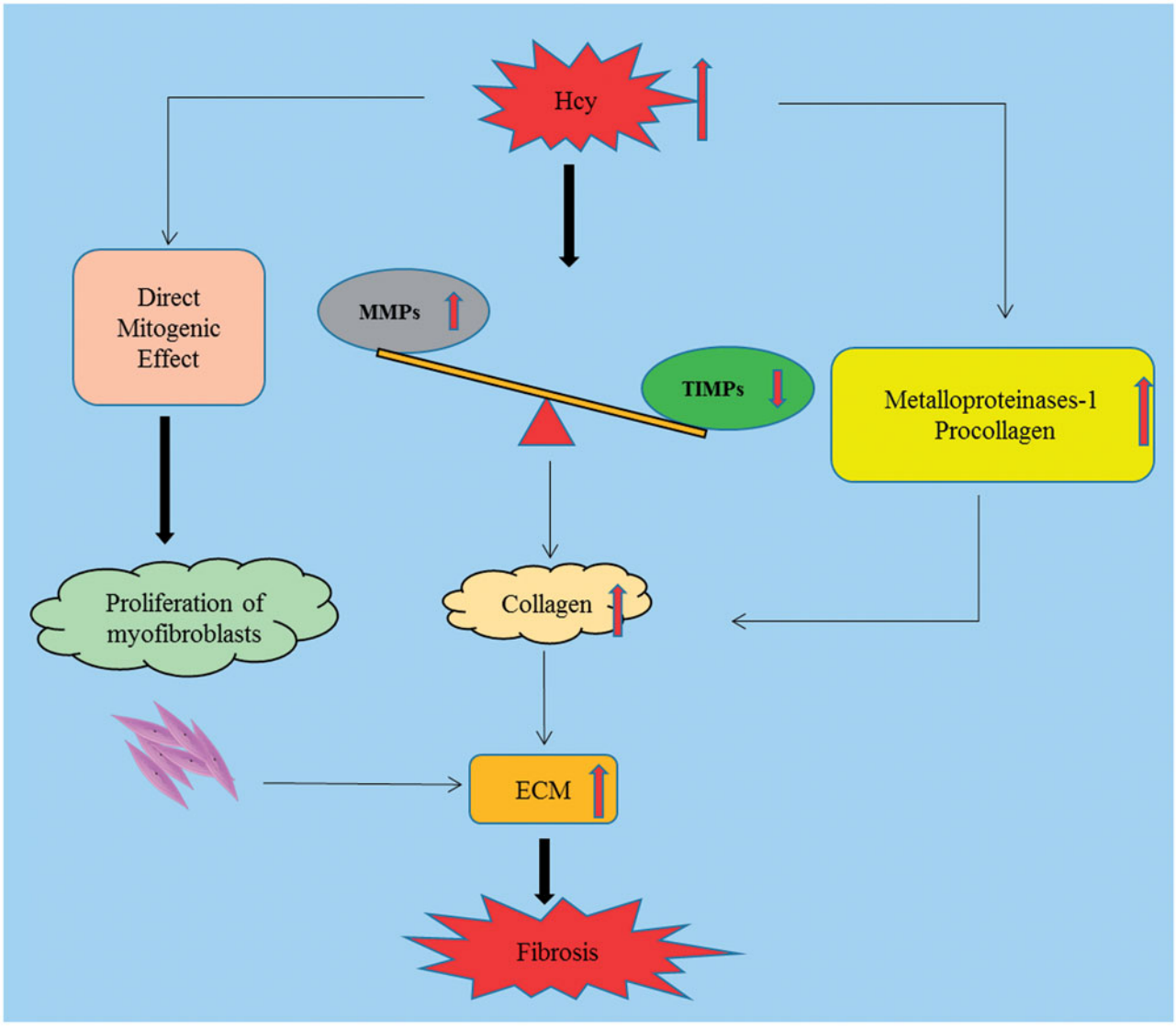

Figure 4. Possible mechanism of Hcy in fibrosis of SSc. Hcy may induce vascular damage through promoting ECM production, influencing MMP/TIMP balance and inducing myofibroblast proliferation and differentiation. MMPs: matrix metalloproteinases; TIMPs: tissue inhibitors of metalloproteinases; ECM: extracellular matrix.

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oxidant status by generating superoxide radicals may exaggerate these pathogenetic mechanisms. Finally, it has been reported that oxidized lipoproteins and the susceptibility to oxidation of low-density lipoproteins were increased in scleroderma [38]. Hcy may have an effect on oxidized lipoproteins and the oxidation of low-density lipoproteins. Therefore, Hcy may play an important role in both macrovasculature and microvasculature involvement in SSc.

6. Possible mechanisms of Hcy in fibrosis of SSc

Fibrosis is the formation of excessive extracellular matrix (ECM) deposition in an organ or tissue during reparative or reactive process. Previous studies have also revealed that elevated plasma levels of Hcy are closely associated with multiple fibrotic diseases including cardiac remodeling, glomerulosclerosis, liver fibrosis and pulmonary fibrosis [39,40]. Skin and organ fibrosis is an important pathological features in SSc, and the epidemiology study has shown that plasma Hcy levels are positively correlated with modified Rodnan total skin score [15]. These suggest that Hcy may be associated with skin fibrosis of SSc. Next, we will focus on the possible mechanisms of Hcy in fibrosis of SSc (Figure 4).

During the process of ECM degradation, matrix metalloproteinases (MMPs) and tissue inhibitors of metalloproteinases (TIMPs) play a pivotal role in this process. MMPs and TIMPs are central to ECM turnover and actively regulate inflammation and remodeling [41]. Hcy seems to regulate the expression and synthesis of tissue inhibitor of metalloproteinases-1 (TIMP-1), and influence MMP/TIMP balance. In liver fibrosis, Torres et al. [42] have found that Hcy is able to induce the expression and synthesis of the TIMP-1 in many different cell types. At the same time, they also found that Hcy could induce the expression of α1(I) procollagen mRNA and promote collagen synthesis in vascular smooth muscle cells [42]. These studies suggest that Hcy may alter extracellular matrix homeostasis. Moreover, in another rat model of intestinal fibrosis, it has been found that the protein expression of TGF-β1, MMP-2, MMP-9 and

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connective tissue growth factor as well as mRNA expression of TGF-β1, MMP-2, MMP-9 and TIMP-1 was significantly increased after injecting Hcy compared with normal group. At the same time, it also found that the expression level of collagen I and collagen III was increased in normal with Hcy injection group when compared with the normal control group [43]. These suggested Hcy promotes intestinal fibrosis which may be attributed to its effects of increasing inflammatory damage, promoting the expression of profibrogenic cytokines and influencing MMP/TIMP balance.

In addition to the regulation of MMPs and TIMPs in fibrosis, Hcy also participates in fibrosis through other mechanism. In liver fibrosis, Hcy is an intermediate in methionine metabolism, which takes place mainly in the liver. Impaired liver function leads to altered methionine and Hcy metabolism. García-Tevijano et al. [44] have shown that the TIMP-1 and α1(I) procollagen were upregulated in vascular smooth muscle cells and liver stellate cells after exposing to pathological concentrations of Hcy. The similar results have also been reported by Sen et al. [45], and they revealed that Hcy induces alpha (α)-smooth muscle actin (a-SMA) and collagen I expression in mouse aortic endothelial cells as evidenced by immunoblot and confocal imaging. Moreover, Lei et al. [46] have demonstrated that elevated concentration of Hcy induced the expression of collagen I in both cultured human liver cells and HHey mice cells. Taken together, HHey may promote ECM production.

7. Conclusion

In this review, we have summarized the latest studies regarding Hcy concentration in SSc and the association between Hcy concentration and clinical features of SSc. Although our preliminary meta-analysis showed that the serum levels of Hcy in SSc were higher than that in healthy controls (with two outlier studies excluded), several limitations need to be acknowledged and these results still need to be demonstrated by well-designed prospective studies whether Hcy was higher in SSc. However, most studies have also shown a higher plasma Hcy level in other systemic rheumatic diseases, and some symptoms of SSc could also be improved significantly by using strategies to lower plasma Hcy level. Therefore, the plasma Hcy level still could be used as a therapeutic target and potential biomarker and can help clinicians in developing new therapeutic strategies and designing novel drugs against HHey. In addition to the investigation of Hcy concentration in SSc, we also summarized the reason why Hcy was higher in SSc. And we found that high Hcy concentration in SSc patients may be associated with folate and vitamin B12 deficiency, MTHFR and CBS gene defects, and some unhealthy lifestyle such as smoking and drinking.

Unfortunately, in this secondary data analysis, some researchers explored the difference in plasma Hcy levels between limited cutaneous SSc and diffuse cutaneous SSc subtypes, while others analyzed the correlation between Hcy levels and clinical features of SSc, and the results were contradictory or inconclusive among these findings. Therefore, we were not able to further illustrate whether the plasma Hcy level is related to the disease activity of SSc due to some detailed information being unavailable. But, there was one study that showed a significant correlation between NVC patterns ('early', 'active' or 'late' pattern) and plasma Hcy levels that progressively increased from the early to the late pattern ($p = .020$) [13], and this suggest that the SSc patients with high disease activity would have high level of Hcy. However, we cannot ignore this important aspect and further studies will be needed to better explore this association.

On the other hand, we also analyzed the possible mechanisms of Hcy in endothelial dysfunction of SSc. Hcy may induce endothelial dysfunction and vascular damage through inducing oxidant stress and inflammatory or prothrombotic conditions as well as altering NO release and increasing collagen production. For the possible mechanism of Hcy in fibrosis of SSc, there are three possible mechanisms including promoting ECM production, influencing MMP/TIMP balance and inducing myofibroblast proliferation and differentiation. However, these mechanisms are still not completely clear. Therefore, the mechanisms of Hcy in fibrosis and vascular damage of SSc are also needed to be confirmed by direct evidences.

Acknowledgements

This work was partly supported by grants from the Anhui provincial major project of quality engineering teaching research in 2016 (code: 2016jyxm0378), Key Project of the Education Department of Anhui Province Natural Science Research (code: KJ2017A164, KJ2017A686), project of visiting study overseas of the excellent youth backbone in the universities of education department of Anhui Province in 2017 (code:gxfx2017008) and Anhui provincial laboratory of population health and major disease screening and diagnosis.

Conflict of interest

None.

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